KONGU ENGINEERING COLLEGE
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DEPARTMENT OF CSE

MACHINE LEARNING BASED DIABETES PREDICTION SYSTEM

24GEP41- MINI PROJECT
FINAL REVIEW

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PROBLEM DESCRIPTION

PROBLEM STATEMENT:

- Diabetes is a **chronic metabolic disease** where **blood sugar levels** are persistently **high** due to **insufficient insulin production**.
- Diabetes affects 830 million people, causing **1.5 million deaths annually.**
- Leads to severe **complications** like **heart disease** and **kidney failure.**
- Current diagnostic methods often **fail to identify high-risk individuals early** enough.

REQUIREMENT:

Urgent need for an efficient, machine learning based predictive tool that **provide early risk assessment** and **supportive preventive healthcare.**

DOMAIN OVERVIEW

CLINICAL SIGNIFICANCE

Early detection is critical to prevent these outcomes.

KEY CLINICAL DIAGNOSTIC MARKERS

The **primary diagnostic thresholds** used by medical professionals are

DIAGNOSTIC MARKER	OPTIMAL RANGE	DIABETICS RANGE
HbA1c (Glycated Haemoglobin)	Below 5.7%	>= 6.5%
Blood Glucose	100 to 125 mg/dL	>= 126 mg/ dL

ROLE OF MACHINE LEARNING

Machine Learning can **identify complex patterns in patient clinical data**, enabling automated and scalable early risk prediction that complements clinical decision-making.

LITERATURE SURVEY

RESOURCE	FOCUS OF THE SOLUTION	MODELS USED	DATASET USED	ACCURACY OBTAINED
International Journal of Advanced Computer Science (2024)	Enhancing diabetes prediction through **advanced feature engineering** (RFECV) and **hyperparameter tuning (**GridSearchCV).	Ensemble model using Random Forest, Gradient Boosting, XGBoost, LightGBM, and CatBoost	PIMA Indians Dataset	94% using XGBoost
MDPI Diagnostics (2023)	Addressing **dataset imbalance** and small sample sizes using **synthetic oversampling** (SMOTE) and dimensionality reduction (PCA).	Bernoulli Naïve Bayes (BNB), Decision Tree, Logistic Regression, and SVM.	PIMA Indians Dataset	79.6% - KNN with SMOTE 77.2% - BNB
International Journal of Scientific Research in CSE & IT (2020)	Early-stage diabetes prediction by **evaluating common classification algorithms**	K-Nearest Neighbor, Logistic Regression, Random Forest, SVM, and Decision Tree	2000-case dataset	99% accuracy – Decision Tree

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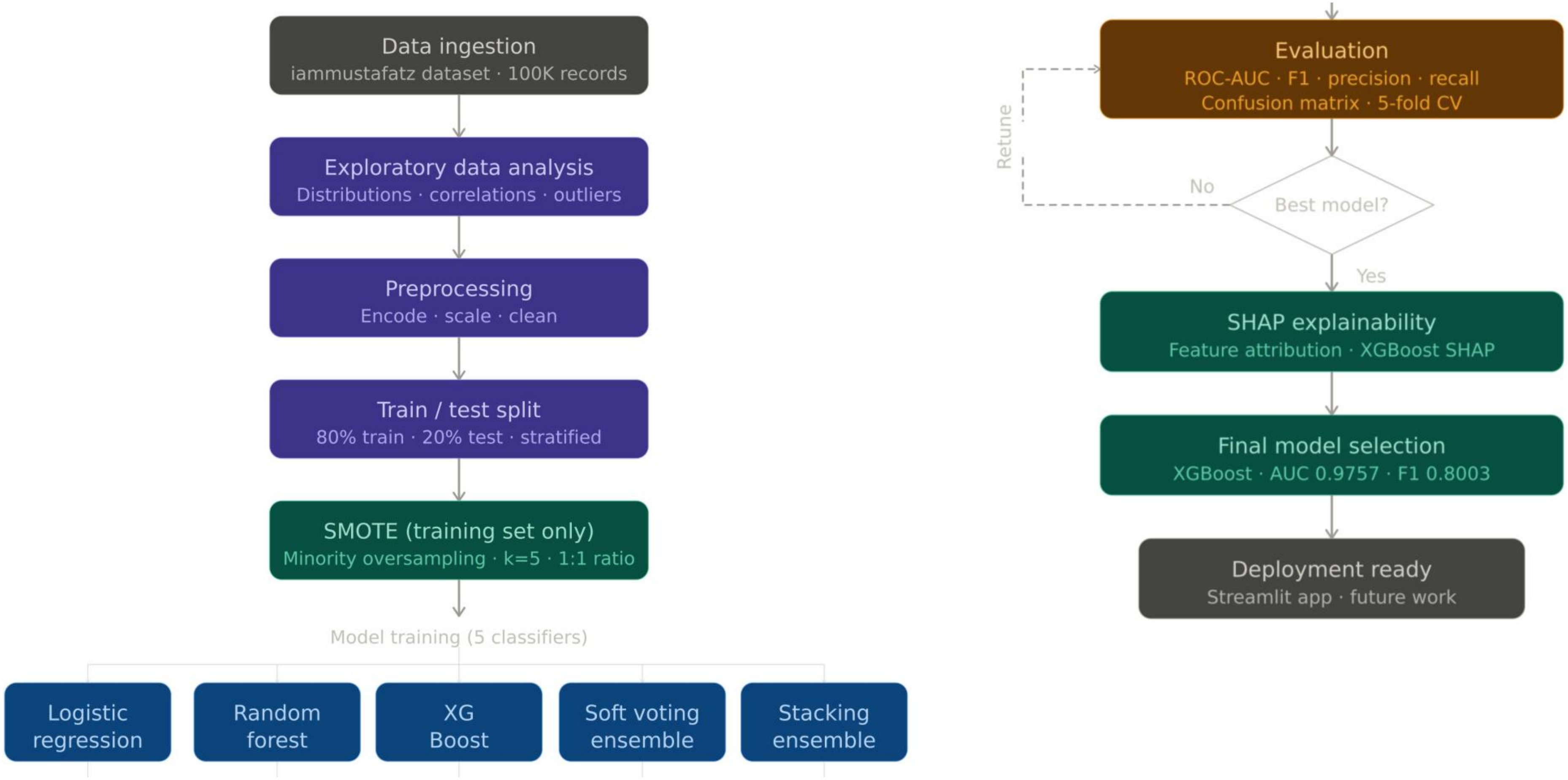
WORKFLOW OF THE PROPOSED SYSTEM



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DATASET DESCRIPTION

- **SOURCE** : Kaggle
- **NO OF DATA:** 100,000 patient records
- **NO OF FEATURES:** 8 input features + 1 target variable

FEATURE	TYPE	RANGE	CLINICAL RELEVANCE
gender	Categorical	Male / Female	Demographic factor
age	Continuous	1 – 80 years	Risk increases with age
hypertension	Binary	0 / 1	Known comorbidity
heart_disease	Binary	0 / 1	Known comorbidity
smoking_history	Categorical	4 categories	Lifestyle risk factor
bmi	Continuous	10.0 – 60.0	Obesity linked to Type 2
HbA1c_level	Continuous	3.5 – 9.0%	**Primary diagnostic marker**
blood_glucose_level	Continuous	80 – 300 mg/dL	**Primary diagnostic marker**
diabetes (target)	Binary	0 / 1	Non-diabetic / Diabetic

WHY THIS DATASET ?

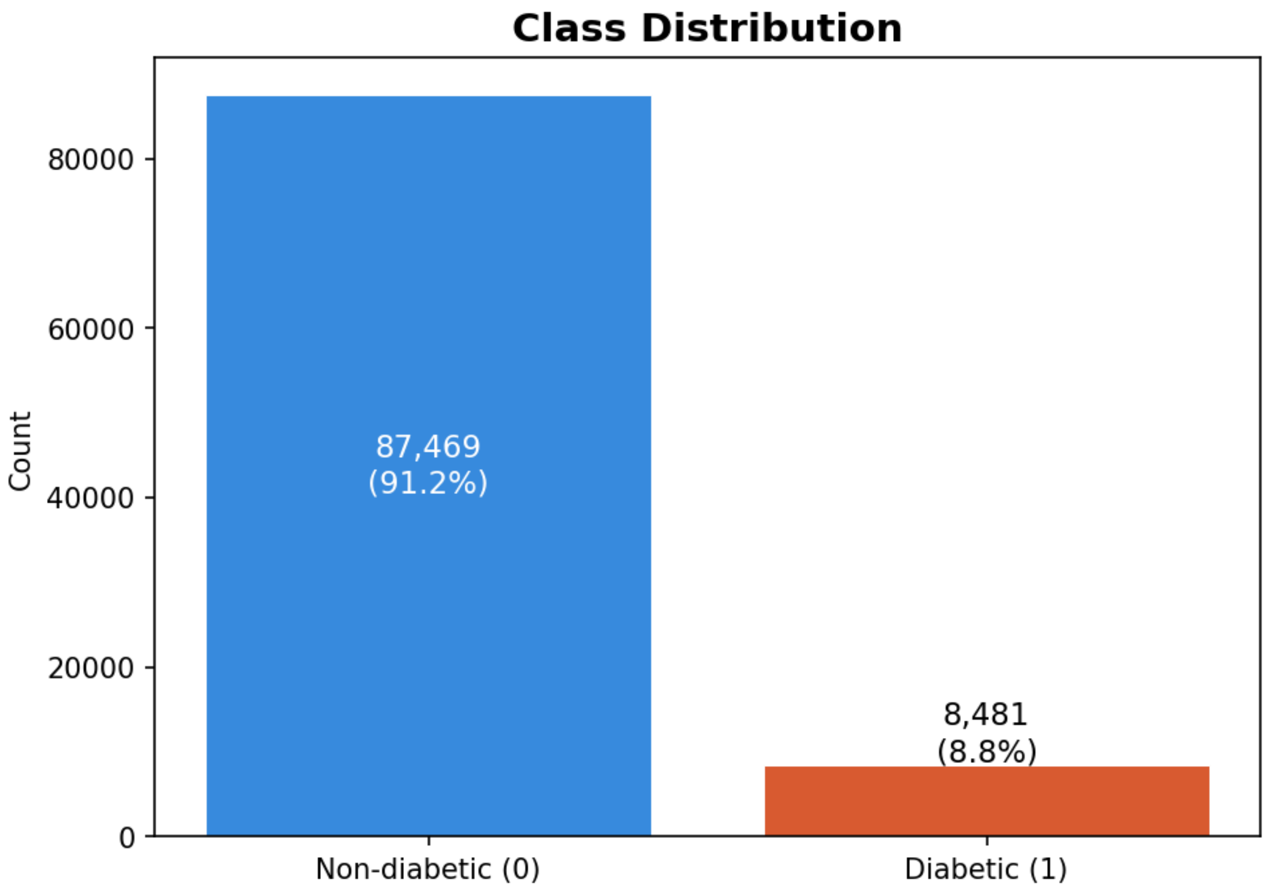
1. **100,000 records** which is **large enough** for **robust model training** and evaluation

2. Contains **real clinical biomarkers - HbA1c and blood glucose** which are the gold standard diagnostic features for diabetes

3. **Chosen over PIMA dataset** which has **only 768 records,** covers only females of one ethnic group, and contains impossible zero values in clinical features

CLASS IMBALANCE IN THE DATASET

- **91,500 Non-diabetic (91.5%)**
- **8,500 Diabetic (8.5%)**
- Severe imbalance addressed using SMOTE on training set only



FEATURE CORRELATION HEATMAP

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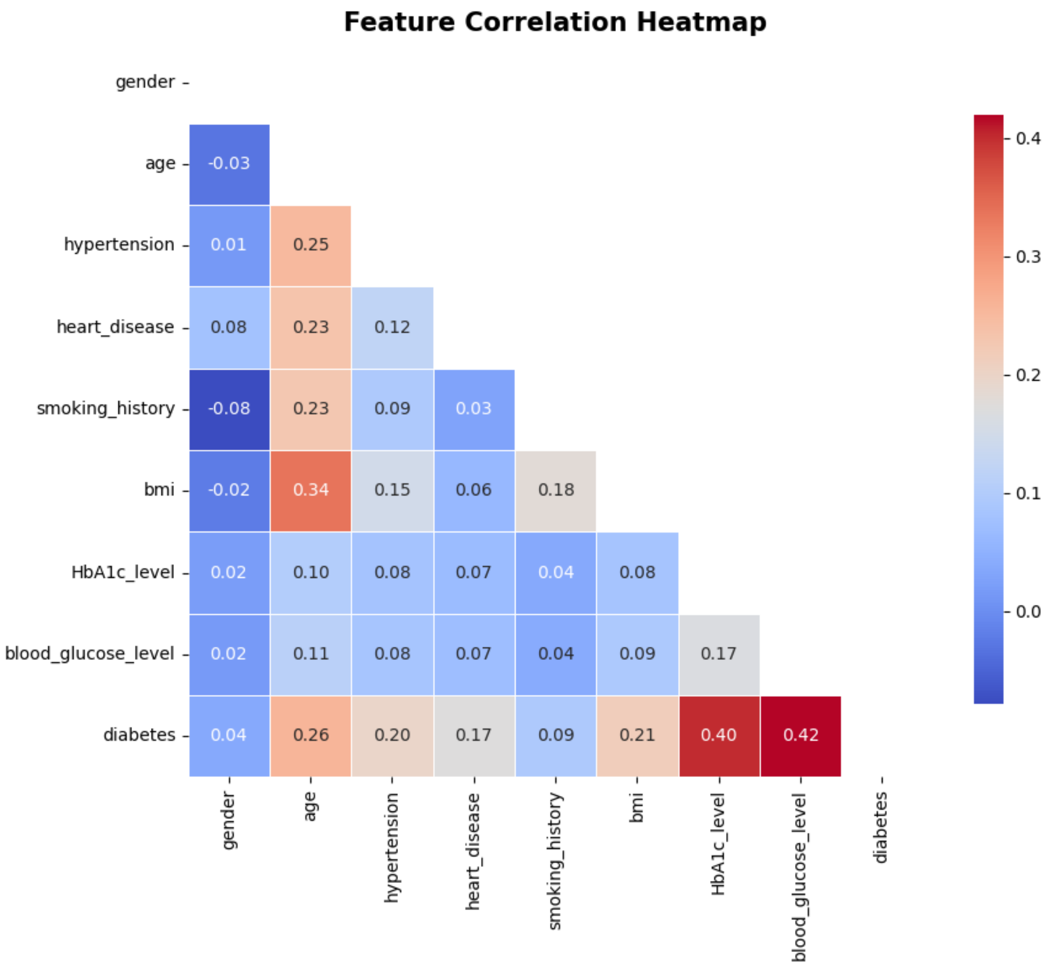
A correlation heatmap is used to visualize the relationship between features in a dataset using Pearson correlation coefficients. H**elps identify the most influential features for diabetes prediction.**

HbA1c level (r = 0.42) showed the **highest positive** correlation with diabetes.

Blood glucose level (r = 0.40) also exhibited a **strong positive** correlation with diabetes.

Age (r = 0.26) and **BMI (r = 0.21)** showed moderate positive correlations.

Gender and **smoking history** had near-zero correlations, indicating weak relationships with diabetes.



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MACHINE LEARNING BASED DIABETES PREDICTION SYSTEM

– Workflow / Pipeline –

1 PHASE 1
EDA (Exploratory Data Analysis)

- Plotted class distribution (8.5% Diabetic, 91.5% Non-Diabetic)
- Plotted histograms for all numerical features
- Correlation heatmap (HbA1c & Blood Glucose showed strongest correlation with target)
- Box plots – clear separation between diabetic and non-diabetic patients for HbA1c and glucose

→

2 PHASE 2
Preprocessing

- Smoking history merged into 4 clean categories (never, former, current, unknown)
- Removed rows with Gender 'Other' (18 rows)
- Dropped all duplicate rows
- Label Encoding applied to categorical features (Gender, Smoking History)
- Standard Scaling applied to all features (mean = 0, std = 1)

→

3 PHASE 3
Train / Test Split + SMOTE

SMOTE

- 80/20 stratified split (preserves class distribution in both sets)
- **SMOTE** applied on training data ONLY (to prevent data leakage)
- Training set before SMOTE: 79,985 rows (8.5% diabetic)
- Training set after SMOTE: 146,374 rows (50% / 50%)
- Test set remains untouched with real-world distribution (8.5% diabetic)

→

4 PHASE 4
Feature Selection

- Preliminary Random Forest model run
- Feature importance scores analyzed
- Correlation of each feature with target checked
- **Decision:** All 8 features retained – clinically justified, no dimensionality issue

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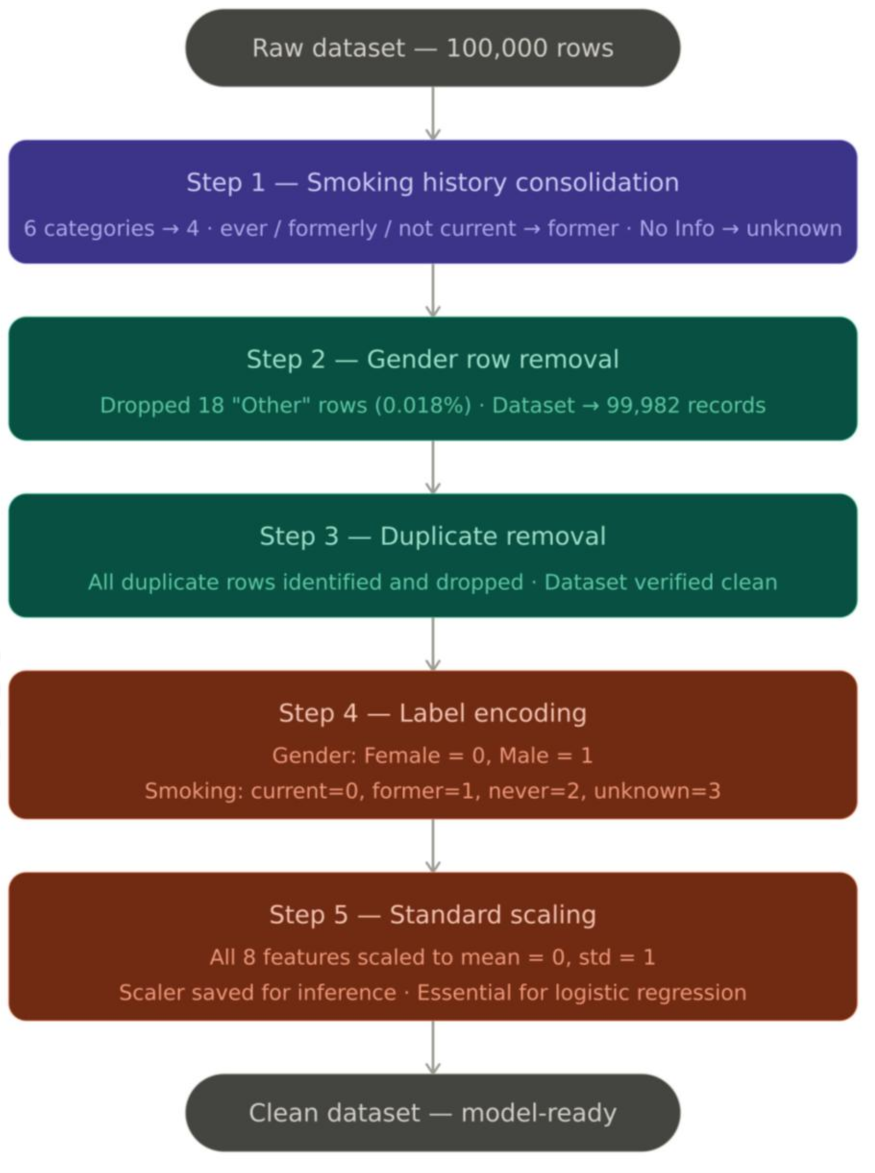
5 PHASE 5
Models Trained

- Logistic Regression (baseline)
- Random Forest (bagging ensemble)
- **XGBoost** (boosting ensemble)
- **Soft Voting** (RF + XGBoost)
- **Stacking** (RF + XGBoost → Logistic Regression meta-model)

OUTPUT
Best performing model: XGBoost
Evaluated using ROC-AUC, F1 Score, Precision, Recall, Confusion Matrix
Explainability using SHAP

DATA PREPROCESSING

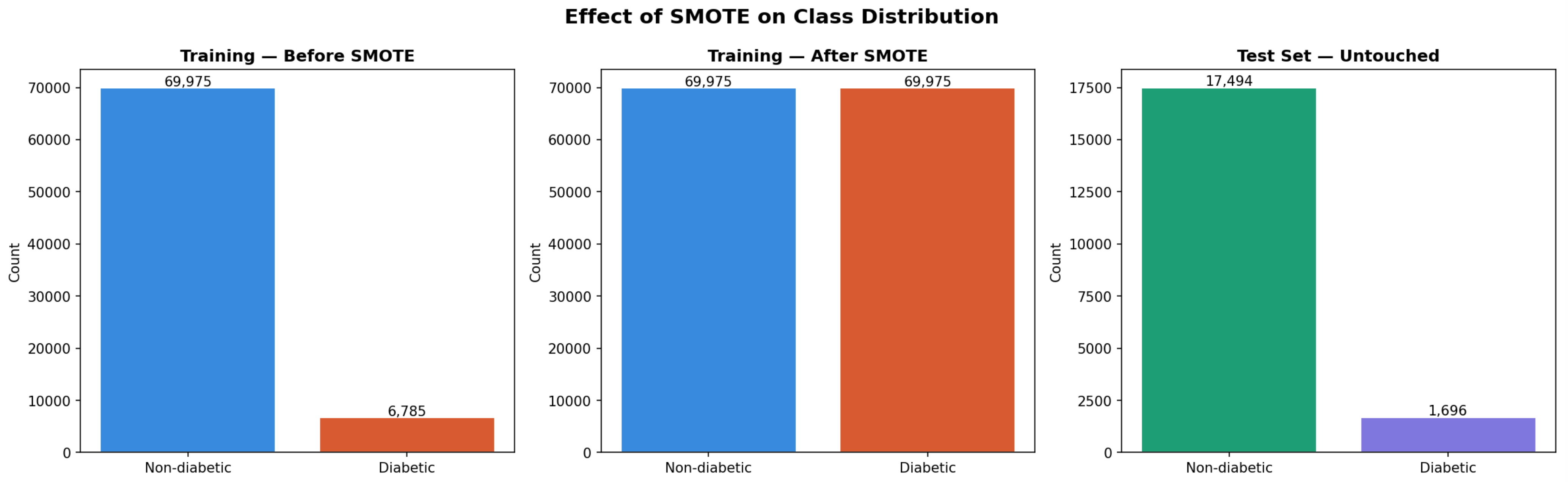


TEST AND TRAIN DATA SPLIT

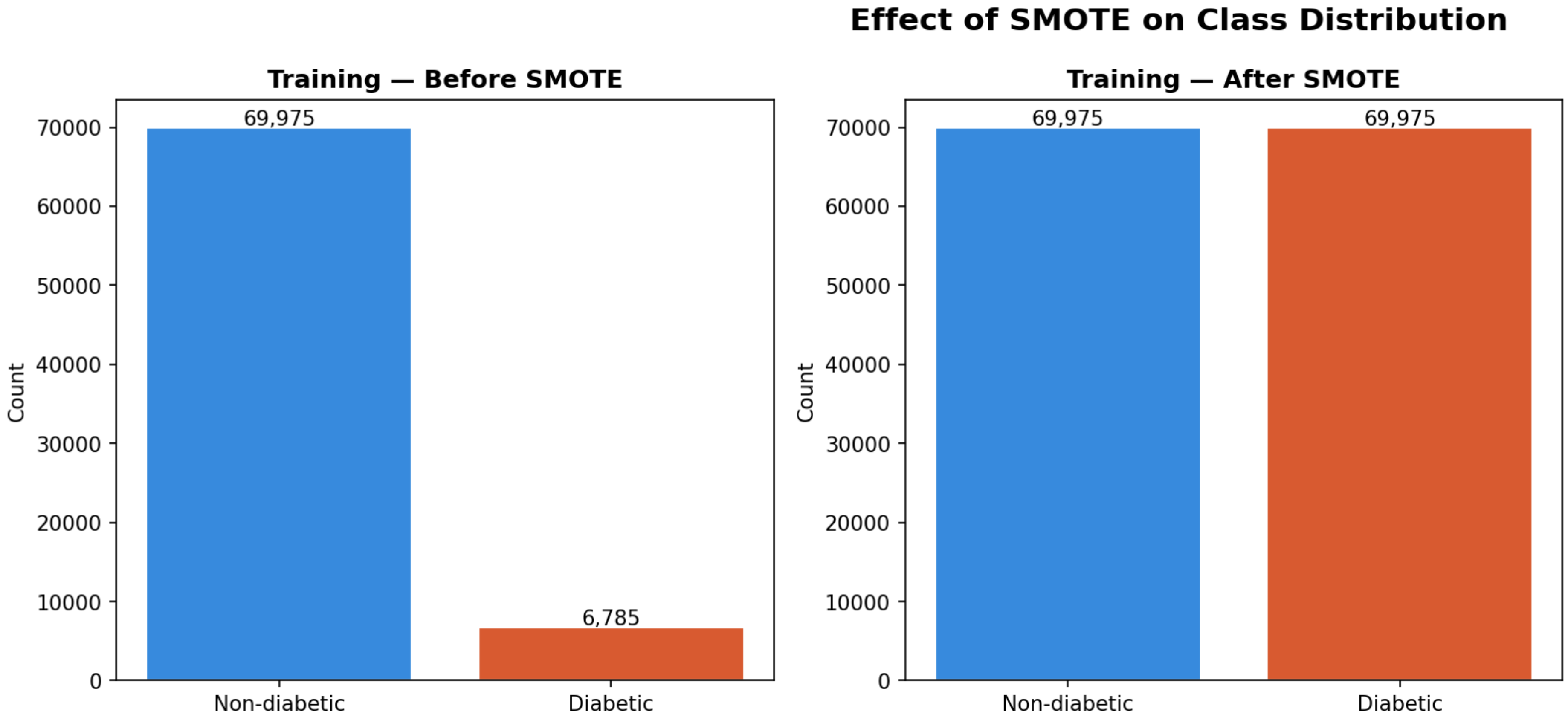
- We split the 99,982 records into **80% for training -** about **79,985** rows and **20% for testing -** about **19,997** rows.

- Then **Synthetic Minority Over-sampling Technique (SMOTE)** was applied on the training set only, growing it from **79,985 rows to 146,374** rows with a perfectly balanced 50/50 class distribution.

- The test set remained completely untouched at its original 8.5% diabetic distribution.



EFFECT OF SMOTE ON TRAINING DATA



EVALUATION METRICS

Model performance is **evaluated using a comprehensive set of metrics** that account for class imbalance and are clinically appropriate for binary medical classification. Accuracy alone is deliberately avoided – a classifier that predicts non-diabetic for every patient would achieve 91.5% accuracy.

Evaluation Metrics used :

1. Confusion Matrix
2. Precision:
3. Recall
4. F1 Score
5. ROC-AUC (Receiver Operating Characteristic - Area Under the Curve)

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MODELS TRAINED

MODEL PERFORMANCE SUMMARY

Model	ROC-AUC (Higher is Better)	F1 Score (Higher is Better)	Precision (Higher is Better)	Recall (Higher is Better)	Remarks
XGBoost (Best Model)	97.57%	80.03%	89.08%	72.64%	Best overall performance
Soft Voting Ensemble (RF + XGBoost)	97.49%	79.03%	84.50%	74.23%	Strong performance
Stacking Ensemble (RF + XGBoost → Logistic Regression)	97.41%	77.75%	80.71%	75.00%	Good balance of precision & recall
Random Forest (Bagging Ensemble)	96.92%	75.87%	75.45%	76.30%	Good recall
Logistic Regression (Baseline Model)	96.16%	57.59%	42.65%	88.62%	High recall, low precision

OVERFITTING CHECK

Train AUC	Test AUC	AUC Gap	5-Fold CV Std Dev (of AUC)	Verdict
99.82%	97.57%	2.25% (Acceptable)	< 2.00% (Stable)	NOT OVERFITTING Model generalizes well

LOGISTIC REGRESSION

```
...
=======================================================
  Logistic Regression
=======================================================
  Training complete.

  ROC-AUC  (Test)  : 0.9616
  ROC-AUC  (Train) : 0.9631
  Gap (Train-Test) : 0.0016
  F1 Score         : 0.5759
  Precision        : 0.4265
  Recall           : 0.8862

  Classification Report:
              precision    recall  f1-score   support

Non-Diabetic       0.99      0.88      0.93     17494
    Diabetic       0.43      0.89      0.58      1696

    accuracy                           0.88     19190
   macro avg       0.71      0.89      0.75     19190
weighted avg       0.94      0.88      0.90     19190
```

RANDOM FOREST

```
...
    ==============================================================
      Random Forest
    ==============================================================
      Training complete.

      ROC-AUC  (Test)  : 0.9692
      ROC-AUC  (Train) : 1.0000
      Gap (Train-Test) : 0.0308
      F1 Score         : 0.7587
      Precision        : 0.7545
      Recall           : 0.7630

      Classification Report:
                    precision    recall  f1-score   support

    Non-Diabetic       0.98      0.98      0.98     17494
        Diabetic       0.75      0.76      0.76      1696

        accuracy                           0.96     19190
       macro avg       0.87      0.87      0.87     19190
    weighted avg       0.96      0.96      0.96     19190
```

XG BOOST

```
========================================================
  XGBoost
========================================================
  Training complete.

  ROC-AUC  (Test)  : 0.9757
  ROC-AUC  (Train) : 0.9982
  Gap (Train-Test) : 0.0225
  F1 Score         : 0.8003
  Precision        : 0.8908
  Recall           : 0.7264

  Classification Report:
              precision    recall  f1-score   support

Non-Diabetic       0.97      0.99      0.98     17494
    Diabetic       0.89      0.73      0.80      1696

    accuracy                           0.97     19190
   macro avg       0.93      0.86      0.89     19190
weighted avg       0.97      0.97      0.97     19190
```

SOFT VOTING (RF + XGBOOST)

```
============================================================
  Soft Voting
============================================================
  Training complete.

  ROC-AUC  (Test)  : 0.9749
  ROC-AUC  (Train) : 1.0000
  Gap (Train-Test) : 0.0250
  F1 Score         : 0.7903
  Precision        : 0.8450
  Recall           : 0.7423

  Classification Report:
              precision    recall  f1-score   support

Non-Diabetic       0.98      0.99      0.98     17494
    Diabetic       0.84      0.74      0.79      1696

    accuracy                           0.97     19190
   macro avg       0.91      0.86      0.89     19190
weighted avg       0.96      0.97      0.96     19190
```

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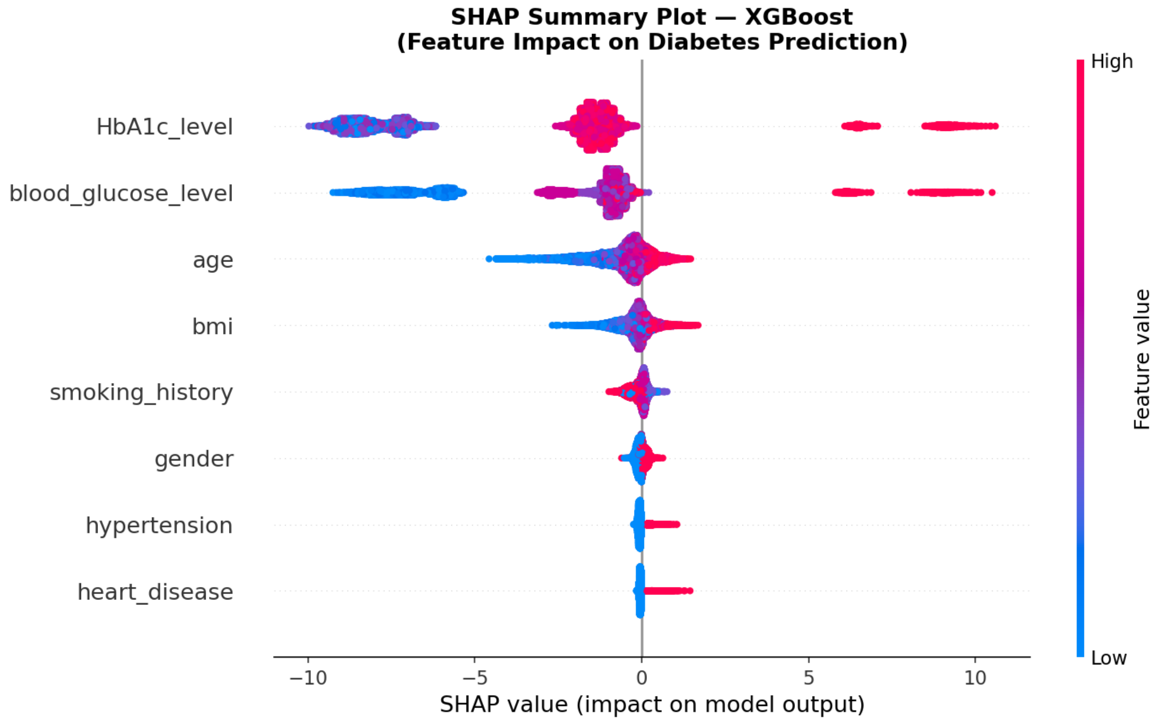
STACKING ENSEMBLE

```
=======================================================
  Stacking
=======================================================
  Training complete.

  ROC-AUC  (Test)  : 0.9741
  ROC-AUC  (Train) : 1.0000
  Gap (Train-Test) : 0.0259
  F1 Score         : 0.7775
  Precision        : 0.8071
  Recall           : 0.7500

  Classification Report:
              precision    recall  f1-score   support

Non-Diabetic       0.98      0.98      0.98     17494
    Diabetic       0.81      0.75      0.78      1696

    accuracy                           0.96     19190
   macro avg       0.89      0.87      0.88     19190
weighted avg       0.96      0.96      0.96     19190
```

SHAP EXPLAINABILITYANALYSIS

- SHAP analysis was performed using **TreeExplainer** on the XGBoost model.
- **HbA1c level** and **blood glucose level** were identified as the most important features for diabetes prediction.
- High values increased the probability of predicting diabetes, while low values reduced it.
- **Age** and **BMI** showed moderate influence on predictions.
- SHAP dependence plots matched clinical diagnostic thresholds such as:
    1. HbA1c ≥ 6.5%
    2. Blood glucose ≥ 140 mg/dL



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THANK YOU